1.10 Multiplying and Dividing Radical Expressions

Lesson Introduction

Benchmark	MA.912.NSO.1.4 Apply previous understanding of operations with rational numbers to add, subtract, multiply, and divide numerical radicals.	Learning Target	• I can multiply and divide two square roots or two cube roots.

	Building On	Working Toward
Benchmark Coherence	MA.8.NSO.1.1 Extend previous understanding of rational numbers to define irrational numbers within the real number system. Locate an approximate value of a numerical expression involving irrational numbers on a number line.	MA.912.NSO.1.5 Add, subtract, multiply and divide algebraic expressions involving radicals.

Essential Questions	• How do I multiply or divide two radical expressions? • When is it appropriate to rationalize the denominator?
Lesson Narrative	In this lesson, students will extend their knowledge of radical expressions to multiply and divide them. Students will get the opportunity to learn and practice this concept with radical expressions presented in multiple forms. Additionally, students will be introduced to the formal definition and process of rationalizing the denominator.

Component Summary	1.10.1 Warm-Up (~5 min)	Radical expression equivalency number puzzle *(SE p. 50)*	1.10.3 Activity (10 min)	Multiplying and dividing radical expressions card activity *(SE p. 53)*
	1.10.2 Guided Instruction (25 min)	Dividing and multiplying radical expressions *(SE p. 51)*	1.10.4 Your Turn! (5 min)	Multiplying and dividing radical expressions practice activity *(SE p. 53)*
	1.10.5 Wrap-Up (~5 min)	Level of confidence self-assessment related to multiplying and dividing two square roots or two cube roots *(SE p. 54)*		

	Glossary	Materials	Additional Preparation
Resources	<u>Key Terms</u>: • Denominator • Rationalizing the Denominator <u>Additional Terms</u>: • Radical Form • Equivalent Expression	• (Optional) Index cards or paper to make passable cards for the 1.10.3 Activity: Pass Around	• Work through each of these problems prior to teaching. Highlight potential misconceptions that students may have to front-load some of your instructional support provided during the lesson.

<table>
<tr>
<td rowspan="4">Teacher Tips</td>
<th>Common Misconceptions</th>
<th>Things to Consider</th>
</tr>
<tr>
<td>Students may forget to rewrite the radical.
Students may not know their factor pairs well enough to simplify the radicals.</td>
<td>1.10.3 Activity requires students to complete a card passing activity. Consider your student learning styles to determine if you would like to prepare the cards ahead of time, use Anchor Charts, or sheets of paper.
Students should not be required to rationalize the denominator. The skill of rationalizing the denominator of an expression with a radical in the denominator was only necessary before students had access to calculators. Knowing that rationalizing the denominator is possible and results in an equivalent expression should be the reason students are asked to rationalize the denominator when a radical is in the denominator of a fraction.</td>
</tr>
<tr>
<th>Literacy and Language Routines</th>
<th>Mathematical Thinking and Reasoning (MTR) Standard</th>
</tr>
<tr>
<td>Poll the Class
1.10.2 Guided Instruction allows students to choose a method and explain why they chose it. Consider leveraging the Poll the Class protocol to poll the class and determine who has chosen the same method, and if any of the students who chose the same method may have different rationale.</td>
<td>MA.K12.MTR.1.1: Participate in Effortful Learning
1.10.3 Activity has students working in groups to complete a step when multiplying or dividing radical expressions. Each student will complete a step and pass it to the next student for review and movement to the next step.</td>
</tr>
<tr>
<td>Differentiate Instruction</td>
<td colspan="2">For 1.10.2 Guided Instruction (question 4), allow for additional physical movement for kinesthetic learners. Consider designating one side of the room as “agree” and one side as “disagree”. Have students choose a side of the room in relation to Jean’s explanation and have a representative from each group explain why they agree or disagree with Jean’s explanation. Repeat this process for Marcus’s explanation as well.

For 1.10.3 Activity, consider having physical cards made for this activity for a hands-on experience for kinesthetic learners. These cards can be made by printing the page and cutting them out or putting the problems directly onto index cards. Another alternative is to put the problems on chart paper and have students rotate to complete the next step of the given problem on the chart.

Review how students are doing in 1.10.4 Your Turn! to determine if additional support may be necessary after this lesson has been completed. This data can also be compiled with their self-reflection data provided in 1.10.5 Wrap-Up.</td>
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<tr>
<td colspan="3">Lesson Notes:</td>
</tr>
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1.10.1 Warm-Up: True Equations

Component Overview: *Students will complete a number puzzle that leverages their knowledge of equivalent radical expressions.*

1.10.1 Warm-Up: True Equations

1. Using the digits 0 to 9, place a digit in each blue box to make both equations true. Each digit can only be used one time, at most.

$$\sqrt{\boxed{1}\,\boxed{8}} = \boxed{3}\,\sqrt{\boxed{2}}$$

$$\sqrt{\boxed{4}\,\boxed{9}} = \boxed{7}$$

Consider reminding students how to approach this type of problem with the use of digits 0 to 9 only once. Remind them of the work they have done with this in prior lessons and to leverage the tools and resources they have at their disposal to help them solve this problem.

Upon completion of the activity, allow students to share their answers to provide them with the correct answer. Survey the students to determine the different methods, strategies, and approaches used to attack this problem. Have students share if they used any of the same strategies. Invest the time in sharing strategies and ways of thinking so that it can support student thinking as they progress through the course.

1.10.2 Guided Instruction: Multiplying and Dividing Radical Expressions

Component Overview: *Students will receive guided instruction on how to divide and multiply radical expressions by rewriting radicals and using the Laws of Exponents.*

1.10.2 Guided Instruction: Multiplying and Dividing Radical Expressions

Recall the ***quotient of powers*** law of exponents.

$$\frac{a^m}{a^n} = a^{m-n}$$

Ariana used the ***quotient of powers*** law of exponents to simplify $\frac{\sqrt{10}}{\sqrt{10}}$. Her work is shown.

$$\frac{\sqrt{10}}{\sqrt{10}} = \frac{10^{\frac{1}{2}}}{10^{\frac{1}{2}}} = 10^{\frac{1}{2}-\frac{1}{2}} = 10^0 = 1$$

When dividing radicals with the same index, the ***quotient law of radicals*** is used to rewrite the expression. Each of the following equivalent expressions demonstrates the quotient law of radicals.

$$\frac{\sqrt{13}}{\sqrt{13}} = \sqrt{\frac{13}{13}} = \sqrt{1} = 1 \qquad \frac{\sqrt{21}}{\sqrt{21}} = \sqrt{\frac{21}{21}} = \sqrt{1} = 1$$

1. Rewrite $\frac{\sqrt[3]{2}}{\sqrt[3]{2}}$ using the quotient law of radicals.

$$\sqrt[3]{\frac{2}{2}} = \sqrt[3]{1} = 1$$

When dividing radical expressions, rewrite radical expressions in equivalent forms and use the quotient law to eliminate radicals where possible.

How are the quotient of powers and the quotient law of radicals similar?

At this point in the lesson, don't introduce the concept of "rationalizing the denominator". This activity is designed to conceptually connect this concept for students to what they have previously learned. At the end of this lesson, students will be introduced to the formal concept of rationalizing the denominator.

1.10.2 Guided Instruction: Multiplying and Dividing Radical Expressions (continued)

2. Write an equivalent expression for $\frac{\sqrt{125}}{\sqrt{245}}$.

$\frac{\sqrt{125}}{\sqrt{245}}$	
$\frac{\sqrt{125}}{\sqrt{245}} = \frac{\sqrt{25 \cdot 5}}{\sqrt{49 \cdot 5}}$	The radicands 125 and 245 are rewritten as the product of a factor and 5.
$\frac{\sqrt{25 \cdot 5}}{\sqrt{49 \cdot 5}} = \frac{5 \cdot \sqrt{5}}{7 \cdot \sqrt{5}}$	The radicals are rewritten using the *product* law of radicals.
$\frac{5 \cdot \sqrt{5}}{7 \cdot \sqrt{5}} = \frac{5}{7} \cdot \frac{\sqrt{5}}{\sqrt{5}}$	Rewrite each perfect square root.
Because $\frac{\sqrt{5}}{\sqrt{5}} = 1$, $\frac{5}{7} \cdot \frac{\sqrt{5}}{\sqrt{5}} = \frac{5}{7}$.	

3. With your partner, rewrite the expression $\frac{\sqrt{275}}{\sqrt{44}}$ using the quotient laws of radicals.

$$\frac{\sqrt{275}}{\sqrt{44}} = \frac{5\sqrt{11}}{2\sqrt{11}} = \frac{5}{2}$$

4. Jean and Marcus were asked to explain how they would multiply $\sqrt{8} \cdot \sqrt{8}$. Their explanations are shown.

Jean's Explanation	I would first rewrite the radical expressions using rational exponents and then use the ***product of powers law of exponents*** to multiply the like bases. $\sqrt{8} \cdot \sqrt{8} = 8^{\frac{1}{2}} \cdot 8^{\frac{1}{2}} = 8^{\frac{1}{2}+\frac{1}{2}} = 8^{\frac{2}{2}} = 8^1 = 8$

When providing instruction during question 2, allow students to respond to the fill-in-the-blank sections within the table. Ask students to explain where they got the value from and how they know it is correct. Once you have reviewed the table, ask students to summarize the process that occurred during the table, using a different example if necessary.

Marcus's Explanation	I would use the ***product law of radicals*** to multiply the radical expressions and then use the law again to rewrite the square root. $\sqrt{8}\cdot\sqrt{8}=\sqrt{8\cdot 8}=\sqrt{64}=8$

Do you agree that both expressions have a value of 8?
Answers will vary. Yes, I agree because I determined the values were the same in my calculator. Yes, because in both methods, the final value is the same.

5. Explain which method you would use to multiply $\sqrt{12}\cdot\sqrt{12}$.
Answers will vary. I would use the product law of radicals because $\sqrt{12}\cdot\sqrt{12}=\sqrt{144}=12$.

6. Multiply the following radical expressions.
$\sqrt{6}\cdot\sqrt{18}$
$6\sqrt{3}$

In cases where there is a fraction with a radical in the **denominator**, mathematicians use a technique called **rationalizing the denominator** to rewrite an expression.

A model of how to rationalize the expression $\frac{30}{\sqrt{2}}$ is shown.

$$\frac{30}{\sqrt{2}}\cdot\frac{\sqrt{2}}{\sqrt{2}}=\frac{30\sqrt{2}}{\sqrt{4}}=\frac{30\sqrt{2}}{2}=15\sqrt{2}$$

To allow for additional physical movement for kinesthetic learners, consider a physical representation of student response. For example, designate one side of the room as "agree" and one side as "disagree." Have students choose a side of the room in relation to Jean's explanation and have a representative from each group explain why they agree or disagree with Jean's explanation. Repeat this process for Marcus's explanation as well.

For a challenge, consider the following additional question:

- *Another student comes along and says they think both Jean and Marcus are wrong. How can you use your calculator to prove that both Jean and Marcus are correct in their final evaluation?*

Have students explain, using their calculator, how they know both are correct.

Students can leave their answer to question 6 as $\sqrt{108}$. Students may also provide other equivalent answers.

Spend a moment connecting what students can do in this lesson with the formal vocabulary of "rationalizing the denominator." If using a word wall or vocabulary space in the classroom, consider adding this phrase and definition to that space.

1.10.3 Activity: Pass Around

Component Overview: *Students will complete a card passing activity where students will begin a multiplication or division of radical expressions, and other students complete their step. Students may also start the problem on a sheet of regular notebook paper and pass along for the same effect. Students will need to be put into groups of four in this activity.*

1.10.3 Activity: Pass Around

In groups of four, each student completes the **first** step of their problem. The students then pass their problem to the student on their right to complete the next step. For each new step, continue passing the problems to the student on the right.

Student A	**Student B**
$\dfrac{\sqrt[3]{200}}{\sqrt[3]{625}}$ $\dfrac{2\sqrt[3]{5}}{5}$	$\sqrt[3]{14} \cdot \sqrt[3]{20}$ $2\sqrt[3]{35}$
Student C	**Student D**
$\sqrt{24} \cdot \sqrt{30}$ $12\sqrt{5}$	$\dfrac{\sqrt{256}}{\sqrt{108}}$ $\dfrac{8\sqrt{3}}{9}$

Consider having physical cards made for this activity for a hands-on experience for kinesthetic learners. These cards can be made by printing the page and cutting them out or putting the problems directly onto index cards. Another alternative is to put the problems on chart paper and have students rotate to complete the next step of the given problem on the chart.

Have students explain their steps to the other group members after writing the next step. Other members of the group should provide feedback and agree or disagree with the student's next step.

For a challenge, consider adding a blank question where each student creates a problem, and the process repeats itself for the newly created question.

1.10.4 Your Turn!

Component Overview: *Students will work independently to practice multiplying and dividing radical expressions. For this activity, students should be released to work independently.*

1.10.4 Your Turn!

1. Write an equivalent expression for each radical expression.

   a. $\sqrt{9} \cdot \sqrt{9}$ b. $\frac{\sqrt[3]{56}}{\sqrt[3]{77}}$ c. $\frac{2\sqrt{22}}{\sqrt{8}}$

   9 $\frac{2\sqrt[3]{121}}{11}$ $\sqrt{11}$

2. Anjelica was asked to rewrite the expression $\frac{\sqrt[3]{264}}{\sqrt[3]{81}}$. Anjelica's work is shown. Write an explanation to Anjelica describing why her problem is incorrect. Then, show how to correctly work the problem.

$$\frac{\sqrt[3]{264}}{\sqrt[3]{81}} = \frac{\sqrt[3]{8 \cdot 33}}{\sqrt[3]{9 \cdot 9}} = \frac{2\sqrt[3]{33}}{9}$$

Anjelica, it looks like you treated the denominator as though it was a square root when it is a cube root.

$$\frac{\sqrt[3]{264}}{\sqrt[3]{81}} = \frac{\sqrt[3]{8 \cdot 33}}{\sqrt[3]{27 \cdot 3}} = \frac{2\sqrt[3]{33}}{3\sqrt[3]{3}} = \frac{2\sqrt[3]{11}}{3}$$

For students who were struggling in the prior activities in this unit, consider pulling into a small group for differentiated support based on what has been observed. Spend more time on question 1 with the small group to allow them additional instruction and practice on multiplying and dividing radical expressions.

Have students share out their methods for question 2. Consider having students come to the board to show their corrected steps. Ask students to articulate the misconception that Anjelica had and how they would approach teaching her how to correct it.

For a challenge, consider having students create an additional problem similar to question 1 and trade with a partner to solve.

1.10.5 Wrap-Up: Reflection

Component Overview: *Students will complete a self-assessment on their level of confidence related to multiplying and dividing two square roots or two cube roots. This activity is designed to be done independently to review confidence levels for each individual student.*

1.10.5 Wrap-Up: Reflection

1. Color in the arrow up to the statement that best describes your current understanding of the following.

   I can multiply and divide two square roots or two cube roots.
   Answers will vary.

I am so confident; I could explain this to someone else!

I can get to the right answer, but I do not understand well enough to explain it yet.

I understand some of this, but I do not understand all of it yet.

I tried hard and I listened, but I am finding this challenging. I will get help by ….

I do not understand any of this yet. I will get help by ….

Student response to these questions can provide an insight into their level of confidence on multiplying and dividing two square roots or two cube roots. This data can be coupled with student performance to help create potential responses for how to better support each individual student.